

EXP 24: Prolog Program for a Dieting System Based on Disease

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Aim:

To develop a Prolog program that suggests appropriate diet recommendations based on a given disease, using a rule-based expert system approach.

Algorithm:

- Step 1: Start the program.
- Step 2: Create a database containing diseases and their diet recommendations.
- Step 3: Accept or select a disease name.
- Step 4: Search the database for the corresponding disease record.
- Step 5: Retrieve the list of recommended diet suggestions.
- Step 6: Display all recommendations related to that disease.
- Step 7: Repeat the process for any other disease if required.
- Step 8: Verify the displayed suggestions.
- Step 9: Stop the program.

Source Code:

% Program to suggest a Dieting System based on Disease

```
diet(diabetes, [  
    'Avoid sugar and sweets',  
    'Eat more leafy vegetables',  
    'Prefer whole grains over refined carbs',  
    'Avoid sugary fruits'  
]).
```

```
diet(hypertension, [  
    'Reduce salt intake',  
    'Eat more potassium-rich foods',  
    'Avoid processed and canned food',  
    'Limit caffeine and alcohol'  
]).
```

```

diet(obesity, [
    'Follow a low-calorie diet',
    'Avoid fried and fatty foods',
    'Increase fiber intake',
    'Exercise along with a balanced diet'
]).

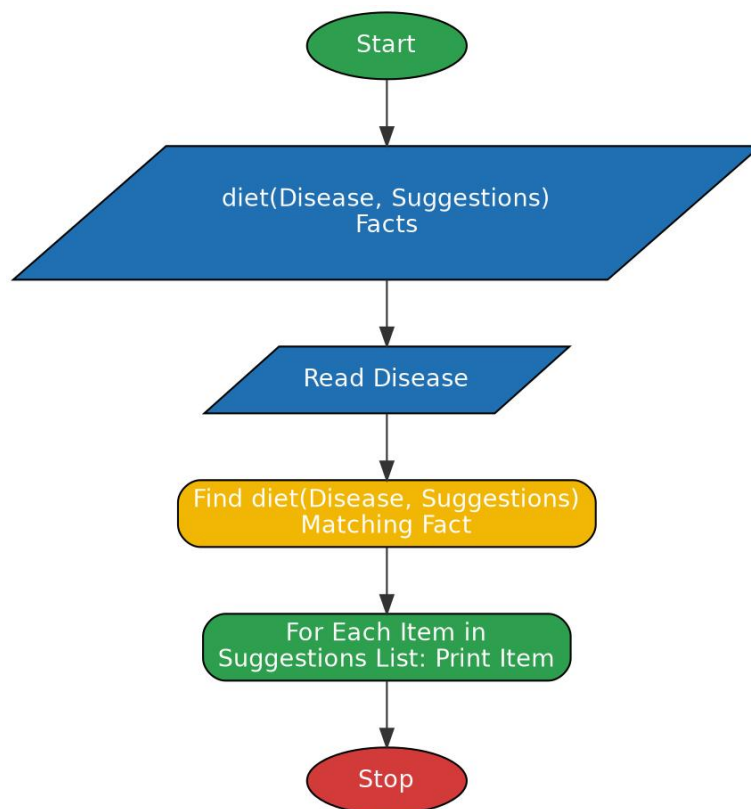
```

```

suggest_diet(Disease) :-
    diet(Disease, Suggestions),
    format('Diet suggestions for ~w:~n', [Disease]),
    forall(member(S, Suggestions),
        format('- ~w~n', [S])).

```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [diet].
true.

?- suggest_diet(diabetes).
Diet suggestions for diabetes:
- Avoid sugar and sweets
- Eat more leafy vegetables
- Prefer whole grains over refined carbs
- Avoid sugary fruits
true.

?- suggest_diet(hypertension).
Diet suggestions for hypertension:
- Reduce salt intake
- Eat more potassium-rich foods
- Avoid processed and canned food
- Limit caffeine and alcohol
true.

?- suggest_diet(obesity).
Diet suggestions for obesity:
- Follow a low-calorie diet
- Avoid fried and fatty foods
- Increase fiber intake
- Exercise along with a balanced diet
true.
```

Result:

Thus the Prolog program for a dieting system based on disease was written, executed successfully and the output was verified.